

Supplementary Materials for
**Human voltage-gated Na⁺ and K⁺ channel properties underlie sustained fast
AP signaling**

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Supplementary Text

Increased excess sodium charge during AP firing in human neurons (Fig S7)

Na⁺ channel inactivation halts sodium influx and prevents excess sodium influx during the falling phase of the AP (22). Excess sodium influx is energetically unfavorable since ion gradients need to be restored at the cost of ATP (22, 39, 40, 59). The reduced and slower inactivation of human Na⁺ currents we observed may therefore come at the cost of more energy use. Given that mouse Na⁺ currents show stronger inactivation and recover slowly from inactivation, we hypothesized that mouse neurons show less excess sodium influx during AP firing (Fig. S7). Indeed, in mouse nucleated patches (n=32), the median excess sodium ratio was 2.3 (1.9-2.7), while in human nucleated patches (n=10) it was 3.8 (3.4-4.1, $p < 10^{-5}$, Wilcoxon rank sum test).

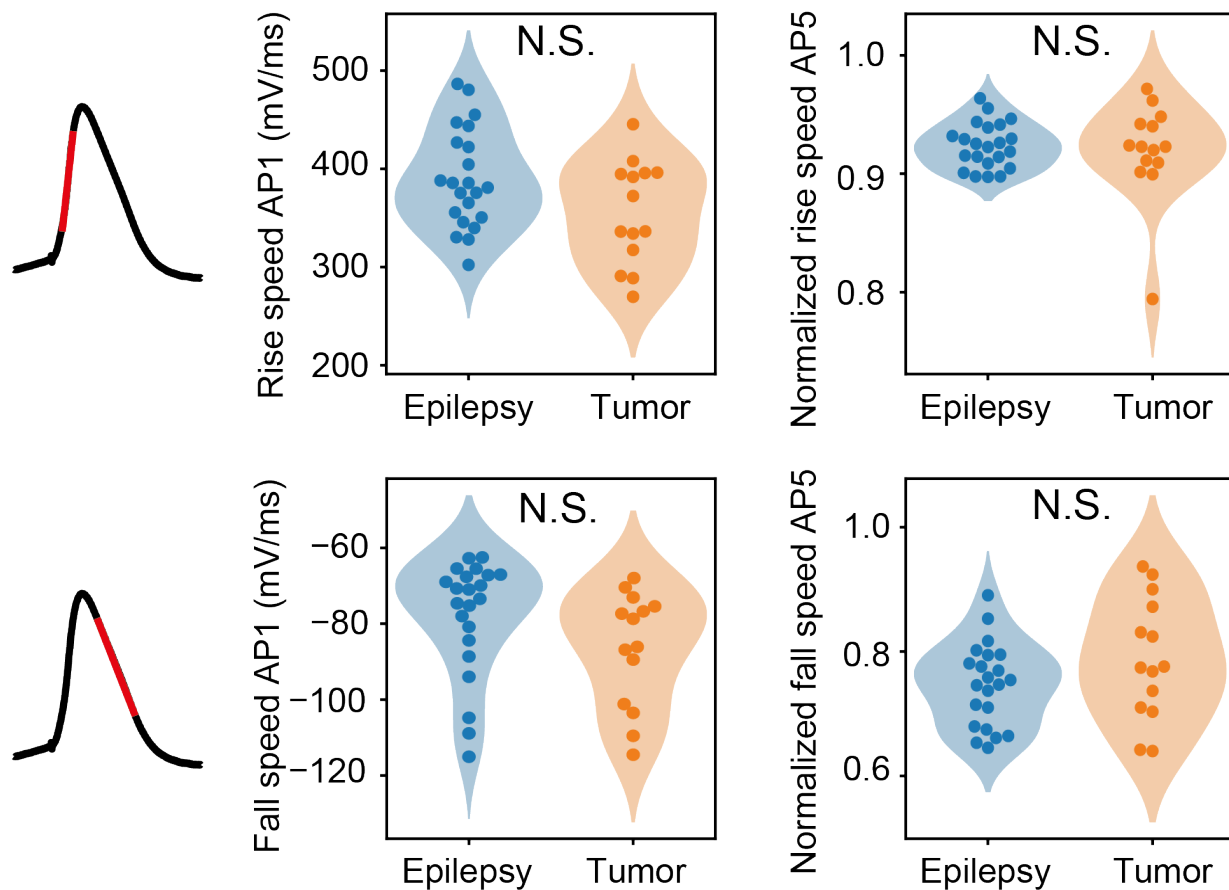


Fig. S1. Stability of AP rise and fall speed is not related to disease history. Top: Rise speed of AP1 (left) and normalized rise speed of AP5 (right) for epilepsy patients vs tumor patients. Bottom: AP1 and AP5 fall speed. Statistical evaluation is indicated (t-test or rank-sum test, after multiple comparison correction). N.S.: $P > 0.05$

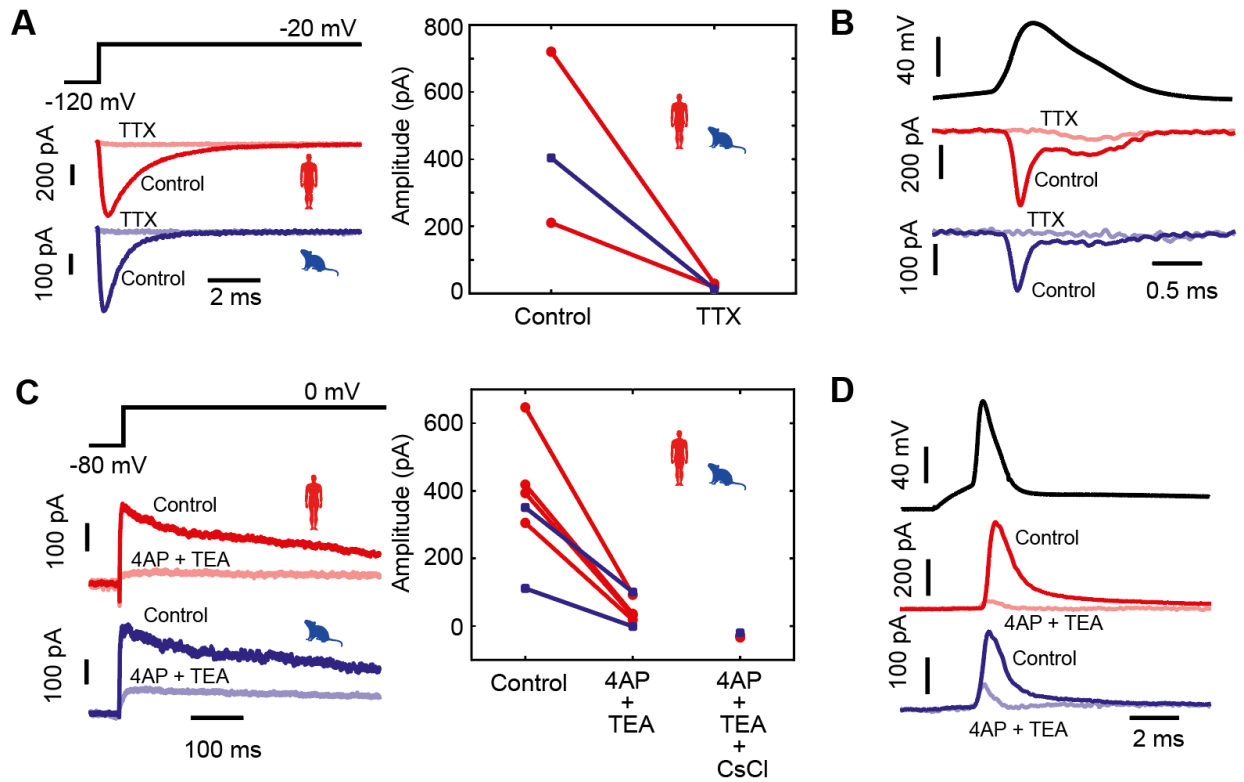


Fig. S2. Pharmacologically isolated voltage-gated Na^+ and K^+ currents. (A) Sodium current example traces before and after application of 1 μ M TTX (left) and plotted amplitudes pre vs post application (right). (B) Sodium currents during AP command voltage-clamp experiment before and after application of 10 μ M TTX. Distinct set of experiments from (A). (C) Potassium current example traces before and after application of 10 mM 4AP + 20 mM TEA (left) and plotted amplitudes pre vs post application (right). Data from distinct experiment shows complete abolishment of 4AP&TEA -insensitive current using CsCl internal. (D) Potassium currents during AP command voltage-clamp experiment before and after application of 10 mM 4AP + 20 mM TEA.

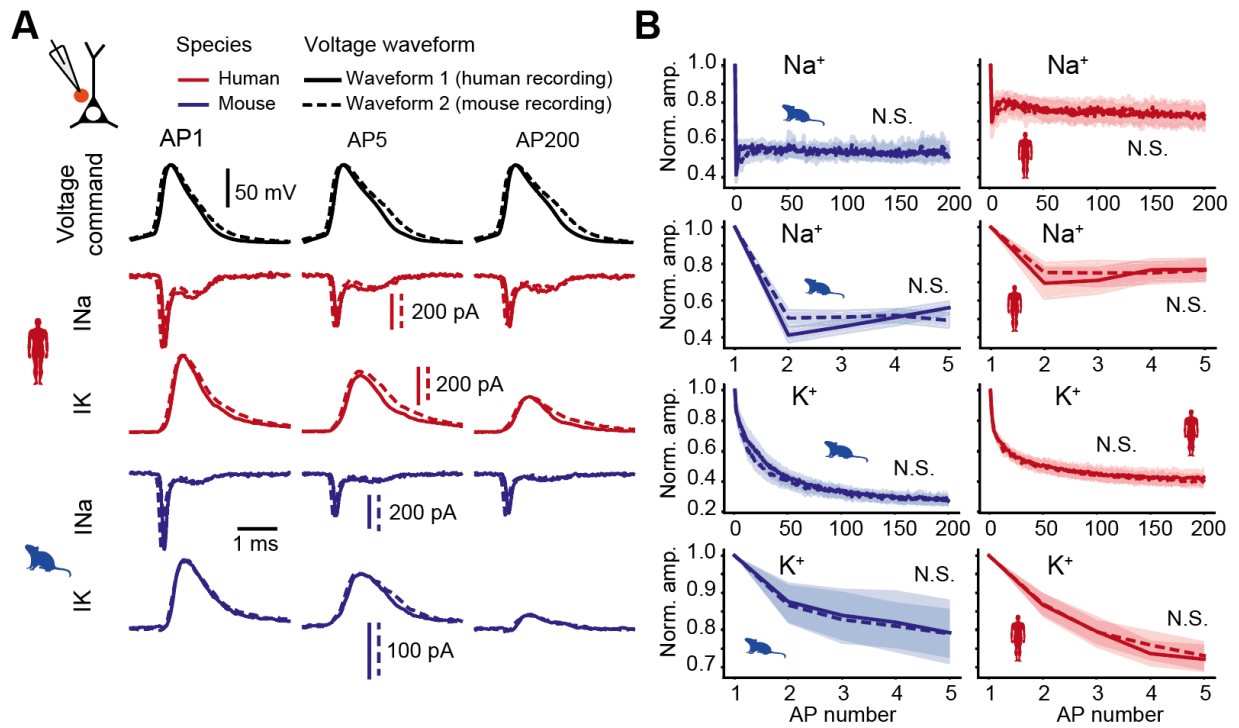


Fig. S3. Species differences in reduction of repetitive Na⁺ & K⁺ currents are not driven by differences in AP command voltage waveform. (A) Similar example traces of voltage-gated Na⁺ and K⁺ currents as in Fig. 2A but with two different AP waveforms. (B) Relative consecutive amplitude of Na⁺ and K⁺ currents for both species during different AP waveforms. Plots are shown for both 200 consecutive APs as well as the first 5 APs in separate plots (data from same experiment). Statistical evaluation of AP waveform was done with multiple linear regression models. N.S.: $p > 0.05$

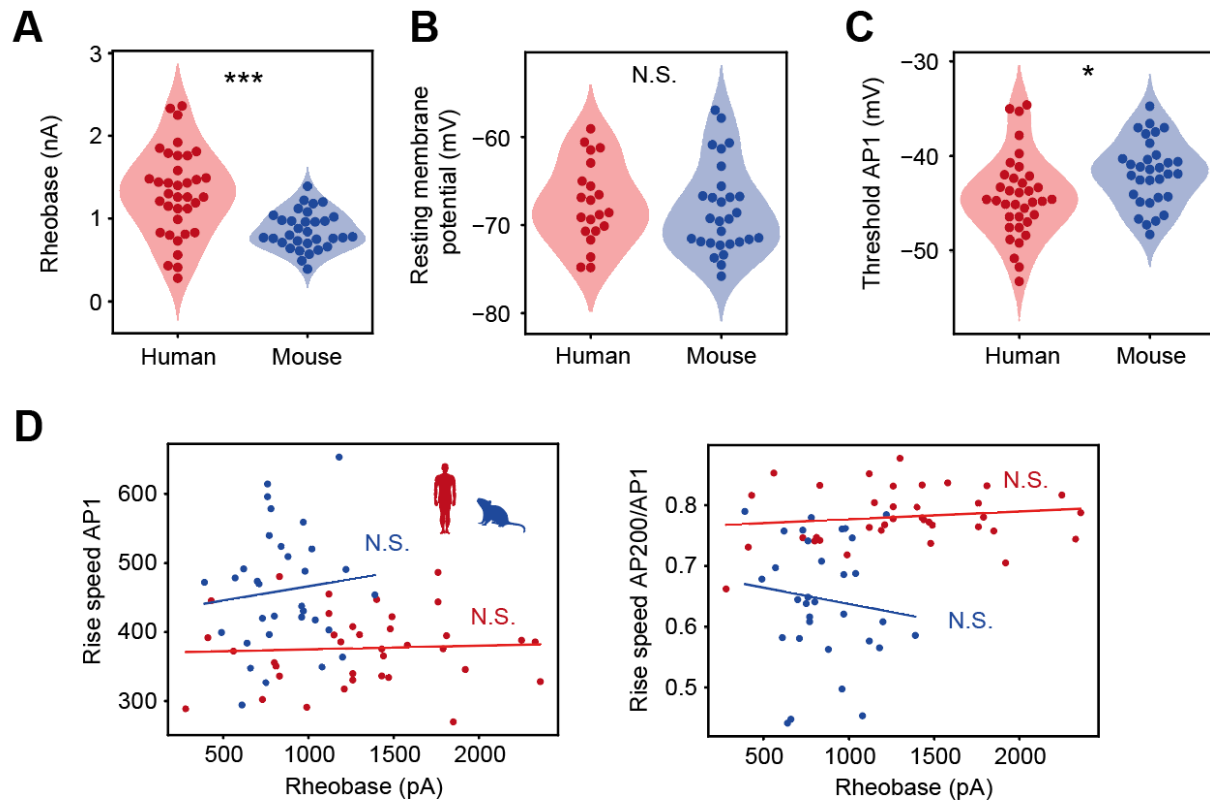


Fig. S4. AP properties in human and mouse neurons. (A) Rheobase, minimum current needed to induced AP firing in response to a 3 ms pulse, was higher in human neurons. T-test, $p=1.9e-05$. (B) Estimated resting membrane potential from neuron's input resistance and injected current to hold the neuron at -70 mV potential. T-test, $p=0.59$. (C) AP threshold was lower in human neurons. T-test, $p=0.011$. (D) Rheobase did not correlate with rise speed of AP1 or normalized rise speed of AP200 in mouse or human neurons. Left: human $R^2=0.003$, $p=0.76$; mouse $R^2=0.012$, $p=0.55$. Right: human $R^2=0.021$, $p=0.4$; mouse $R^2=0.015$, $p=0.51$.

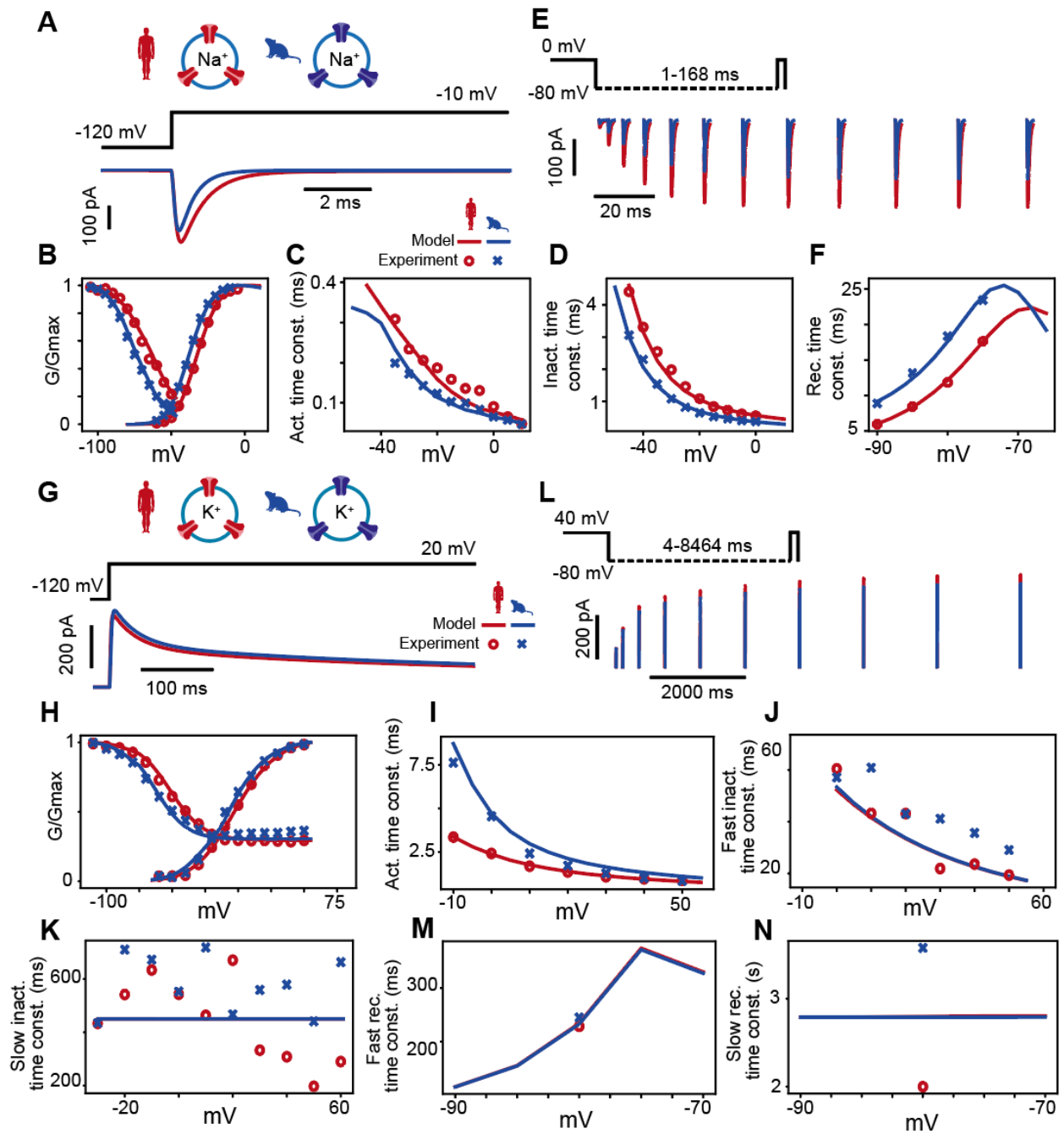


Fig S5. Models of human and mouse Na⁺ & K⁺ channels. (A) Examples of model Na⁺ current responses to a -10 mV voltage. (B) Steady state responses of the Na⁺ models and experimental data are shown. (C) Activation time constants of the experimentally recorded and model Na⁺ currents. (D) Inactivation time constants of the experimentally recorded and model Na⁺ currents. (E) Examples of model recovery Na⁺ currents after variable time delays. (F) Recovery time constants of the experimentally recorded and model Na⁺ currents. (G) Examples of model K⁺

current responses to a -10 mV voltage. (H) Steady state responses of the K^+ models and experimental data. (I) Activation time constants of the experimentally recorded and model K^+ currents. (J) Fast inactivation time constants of the experimentally recorded and model K^+ currents. (K) Slow inactivation time constants of the experimentally recorded and model K^+ currents. (L) Examples of model recovery K^+ currents after variable time delays. (M) Fast recovery time constants of the experimentally recorded and model K^+ currents. (N) Slow recovery time constants of the experimentally recorded and model K^+ currents. Markers represent means of human (o) and mouse (x) experimental data.

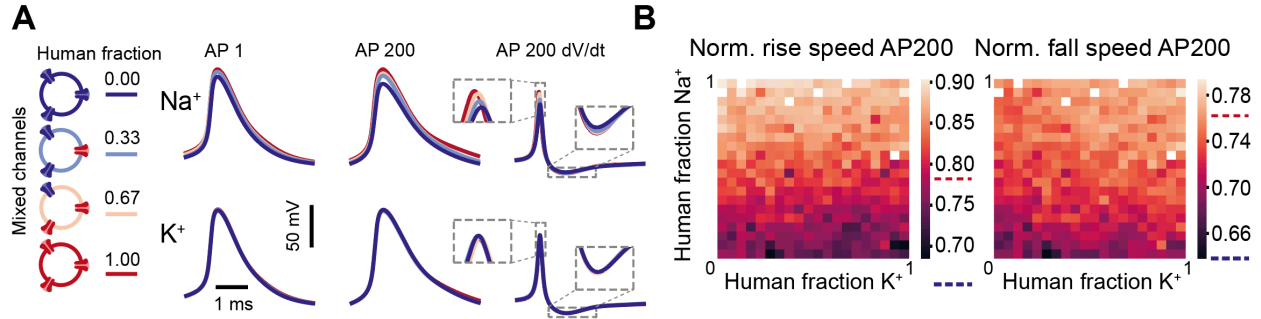


Fig S6. Human-like Na⁺ and K⁺ channel kinetics stabilize APs across 200 APs (A) Active AP firing model with variable fractions of human and mouse Na⁺ and K⁺ channels. Example traces of 1st AP, 200th AP and derivative of the 200th AP for various human proportions for Na⁺ (top) and K⁺ (bottom) channel simulations. (B) Quantification of rise (left) and fall (right) speed of AP200 relative to AP1 and dependence on fraction of Na⁺ and K⁺ channel models with human properties. Conductance densities as in Fig. 6I-J, with matched best-fit densities from Fig. 6G. Dotted lines next to color legends indicate experimental values for mouse (blue) and human (red) relative rise and fall speed of AP200 at 40 Hz; rise speed: human = 0.78, mouse = 0.65; fall speed: human = 0.76, mouse = 0.64.

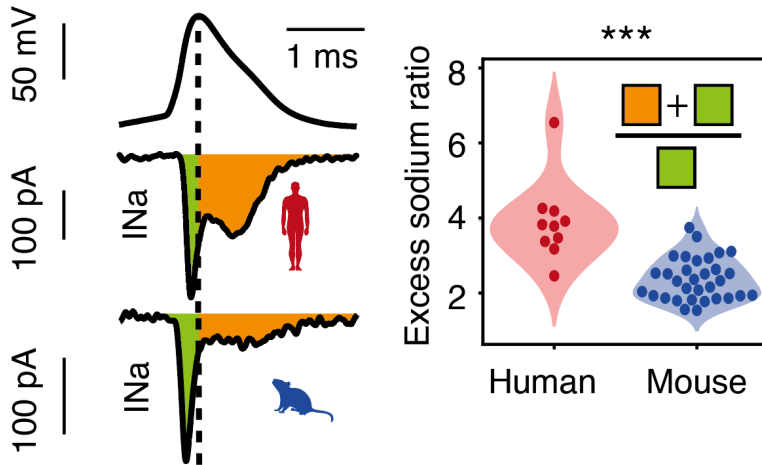


Fig. S7. Increased excess sodium charge during AP firing in human neurons. Left: Sodium current traces during AP voltage waveforms in nucleated patches of human and mouse pyramidal neurons. Green and orange shadings indicate the sodium charge before and after the AP peak. Right: Violin plot of excess sodium ratio in both species (human: n=10 recordings, mouse: n=32 recordings). *** $p < 10^{-5}$, Wilcoxon rank sum test.

Supplementary tables

Table S1. Boltzmann fit characteristics for maximal Na⁺ and K⁺ conductances. Data are not corrected for 15 mV liquid junction potential.

	Human		Mouse		Statistics		
	mean±SD	n	mean±SD	n	Test	Statistic	p-value
Na⁺ activation							
V _{1/2} (mV)	-32.13±4.86	19	-37.88±4.58	15	t-test	3.513	0.0013
Slopefactor (ms ⁻¹)	5.77±0.52	19	5.82±0.43	15	t-test	-0.332	0.74
Na⁺ inactivation							
V _{1/2} (mV)	-66.00±7.82	18	-74.82±6.81	15	t-test	3.416	0.0018
Slopefactor (ms ⁻¹)	7.99±1.92	18	7.3±0.86	15	t-test	1.373	0.18
K⁺ activation							
V _{1/2} (mV)	-3.03±5.95	13	-7.91±6.04	29	t-test	2.432	0.02
Slopefactor (ms ⁻¹)	13.87±1.35	13	13.68±2.17	29	t-test	0.290	0.77
K⁺ inactivation							
V _{1/2} (mV)	-52.69±5.97	11	-65.03±5.32	26	t-test	6.218	4*10 ⁻⁷
Slopefactor (ms ⁻¹)	12.8±1.87	11	11.52±1.97	26	t-test	1.830	0.76

Table S2. Parameters used in NEURON active model.

	Human	Mouse	
General parameters			
Diameter (μm)	10	10	Diameter of the single compartment model
C_m ($\mu\text{F} \cdot \mu\text{m}^{-2}$)	1	1	Specific membrane capacitance
R_a ($\Omega \cdot \text{cm}$)	200	200	Axial resistance
R_m ($\Omega \cdot \text{cm}^{-2}$)	10000	10000	Specific membrane resistance
g_{pas} ($\text{S} \cdot \text{cm}^{-2}$)	0.0001	0.0001	Passive conductance density
e_{pas} (mV)	-85	-85	Reversal potential of passive conductance
$\bar{g}_{\text{Na}}$ ($\text{pS} \cdot \mu\text{m}^{-2}$)	114.5	148.6	Maximal Na^+ conductance density
e_{Na} (mV)	53	53	Na^+ reversal potential
$\bar{g}_{\text{K}}$ ($\text{pS} \cdot \mu\text{m}^{-2}$)	29.05	65	Maximal K^+ conductance density
e_{K} (mV)	-101	-101	K^+ reversal potential
Na^+ channel model			
th_act_inf (mV)	-42.291457	-46.6348	V_{half} for activation steady state
$q_{\text{act_inf}}$ (mV)	10.2311529	10.94042	Slope factor for activation steady state
tha (mV)	-58.592356	-53.6102	V_{half} for (de)activation time constant
q_a (mV)	9.21455372	10.08771	(De)activation time constant slope
R_a (ms^{-1})	0.21398493	0.25131	Open/close rate constant
th_inact_inf (mV)	-64.48	-73.97	V_{half} for inactivation steady state
$q_{\text{inact_inf}}$ (mV)	11.02	9.235	Slope factor for inactivation steady state
thi1 (mV)	-39.298677	38.18728	V_{half} for inactivation time constant
thi2 (mV)	-77.166745	73.39262	V_{half} for recovery from inactivation time constant
$qi1$ (mV)	7.21636387	7.262844	Inactivation time constant slope
$qi2$ (mV)	4.74945005	2.771082	Recovery from inactivation time constant slope
R_d (ms^{-1})	0.04763919	0.078592	Inactivation rate constant
R_g (ms^{-1})	0.01195166	0.006373	Recovery from inactivation rate constant
q_{10}	2.3	2.3	Temperature sensitivity constant
K^+ channel model			
th_act_inf (mV)	-18.406164	-24.9224	V_{half} for activation steady state
$q_{\text{act_inf}}$ (mV)	19.1995503	17.95853	Slope factor for activation steady state
tha (mV)	-29.961798	-17.1079	V_{half} for (de)activation time constant
q_a (mV)	6.59040569	3.688419	(De)activation time constant slope
R_a (ms^{-1})	0.01845925	0.0167	Open/close rate constant
th_inact_inf (mV)	-52.61	-65.2	V_{half} for inactivation steady state
$q_{\text{inact_inf}}$ (mV)	13.09	12.1	Slope factor for inactivation steady state
thi1 (mV)	40.5659149	40.56591	V_{half} for inactivation time constant
thi2 (mV)	-75.938227	-75.9382	V_{half} for recovery from inactivation time constant
$qi1$ (mV)	26.7739292	26.77393	Inactivation time constant slope
$qi2$ (mV)	0.40183383	0.401834	Recovery from inactivation time constant slope
R_d (ms^{-1})	0.00169014	0.00169	Inactivation rate constant
R_g (ms^{-1})	0.00049688	0.000497	Recovery from inactivation rate constant
$h2\tau_{\text{inact}}$ (ms)	450	450	Slow inactivation time constant
$h2\tau_{\text{inact_rec}}$ (ms)	2789.79294	2789.793	Slow inactivation recovery time constant
prop_inact	0.85	0.85	Fraction of inactivating conductance
prop_h2	0.4	0.4	Fraction of slow-inactivating conductance (of prop_inact)
q_{10}	2.3	2.3	Temperature sensitivity constant

125 **Table S3. Surface area and maximal conductance of nucleated patches for Na⁺ and K⁺ current recordings.**

	Human mean±SD	Mouse mean±SD
Na⁺ current recordings		
Surface area (μm ²)	368±63	254±32
Conductance Na ⁺ (pS)	2074±1134	3432±1292
Na ⁺ Conductance density (pS* μm ⁻²)	5.85±3.38	13.39±4.54
K⁺ current recordings		
Surface area (μm ²)	307±63	228±38
Conductance K ⁺ (pS)	9398±4399	9481±2929
K ⁺ conductance density (pS* μm ⁻²)	31.22±14.71	43.47±15.04